BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: C Batch: 3 Practical No: 9

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Subject Name : Deep Learning

Problem Statement : Perform image classification using a pre-trained Vision Transformer (ViT) model and compare its performance with a Convolutional Neural Network (CNN) model using the Oxford-IIIT Pet dataset.

Algorithm :

1. The required libraries, including PyTorch, torchvision, NumPy, Matplotlib, and scikit-learn, are imported.
2. The Oxford-IIIT Pet dataset containing images of 37 cat and dog breeds is loaded.
3. The images are resized to 224 × 224 pixels, converted into tensors, and normalized using ImageNet statistics.
4. Balanced training, validation, and testing subsets are created with 20, 5, and 5 images per class, respectively.
5. The ImageNet-pretrained ResNet50 CNN and ViT-B/16 Vision Transformer models are loaded.
6. The pretrained parameters are frozen, and the original classification heads are replaced with new layers for 37 output classes.
7. The new classification heads are trained for two epochs using the Adam optimizer and cross-entropy loss.
8. Both models are evaluated using accuracy, macro precision, macro recall, macro F1-score, and test loss.
9. Performance comparison graphs, training and validation accuracy curves, and confusion matrices are generated.
10. Finally, the performance and training time of both models are compared, and conclusions are drawn from the results.

Code :

```
!pip install -q torch torchvision scikit-learn matplotlib seaborn
import os,time,random,copy,gc
import numpy as np
import torch
import torch.nn as nn
import torch.optim as optim
import torchvision.models as models
import torchvision.transforms as transforms
import torchvision.datasets as datasets
import matplotlib.pyplot as plt
import seaborn as sns
from torch.utils.data import Dataset,DataLoader,Subset
from sklearn.metrics import accuracy_score,precision_score,recall_score,f1_score,confusion_matrix,classification_report
SEED=42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
if torch.cuda.is_available(): torch.cuda.manual_seed_all(SEED)
IMAGE_SIZE=224
BATCH_SIZE=64
EPOCHS=2
LEARNING_RATE=0.001
TRAIN_PER_CLASS=20
VAL_PER_CLASS=5
TEST_PER_CLASS=5
NUM_CLASSES=37
DEVICE=torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Device:",DEVICE)
if torch.cuda.is_available(): print("GPU:",torch.cuda.get_device_name(0))
```

```
if torch.cuda.is_available(): print("GPU:",torch.cuda.get_device_name(0))
train_transform=transforms.Compose([transforms.Resize((IMAGE_SIZE,IMAGE_SIZE)),transforms.RandomHorizontalFlip(),transforms.ToTensor(),transforms.Normalize([0.485,0.456,0.406]
eval_transform=transforms.Compose([transforms.Resize((IMAGE_SIZE,IMAGE_SIZE)),transforms.ToTensor(),transforms.Normalize([0.485,0.456,0.406],[0.229,0.224,0.225])])
train_raw=datasets.OxfordIIITPet(root="./data",split="trainval",target_types="category",download=True)
test_raw=datasets.OxfordIIITPet(root="./data",split="test",target_types="category",download=True)
```

```
class PetDataset(Dataset):
    def __init__(self,base,indices,transform):
        self.base=base
        self.indices=indices
        self.transform=transform
    def __len__(self): return len(self.indices)
    def __getitem__(self,i):
        image,label=self.base[self.indices[i]]
        return self.transform(image),label
def get_labels(dataset):
    if hasattr(dataset,"_labels"): return list(dataset._labels)
    if hasattr(dataset,"targets"): return list(dataset.targets)
    return [dataset[i][1] for i in range(len(dataset))]
def balanced_indices(labels,count):
    grouped={c:[] for c in range(NUM_CLASSES)}
    for i,label in enumerate(labels):
        grouped[int(label)].append(i)
    selected=[]
    for c in range(NUM_CLASSES):
        random.shuffle(grouped[c])
        if len(grouped[c])<count: raise ValueError(f"Not enough images for class {c}")
        selected.extend(grouped[c][:count])
    return selected
train_labels=get_labels(train_raw)
test_labels=get_labels(test_raw)
grouped={c:[] for c in range(NUM_CLASSES)}
for i,label in enumerate(train_labels): grouped[int(label)].append(i)
train_idx=[]
val_idx=[]
```

```
for c in range(NUM_CLASSES):
    random.shuffle(grouped[c])
    train_idx.extend(grouped[c][:TRAIN_PER_CLASS])
    val_idx.extend(grouped[c][TRAIN_PER_CLASS:TRAIN_PER_CLASS+VAL_PER_CLASS])
test_idx=balanced_indices(test_labels,TEST_PER_CLASS)
train_ds=PetDataset(train_raw,train_idx,train_transform)
val_ds=PetDataset(train_raw,val_idx,eval_transform)
test_ds=PetDataset(test_raw,test_idx,eval_transform)
pin=DEVICE.type=="cuda"
train_loader=DataLoader(train_ds,batch_size=BATCH_SIZE,shuffle=True,num_workers=2,pin_memory=pin)
val_loader=DataLoader(val_ds,batch_size=BATCH_SIZE,shuffle=False,num_workers=2,pin_memory=pin)
test_loader=DataLoader(test_ds,batch_size=BATCH_SIZE,shuffle=False,num_workers=2,pin_memory=pin)
print("Classes:",NUM_CLASSES)
print("Train:",len(train_ds),"Validation:",len(val_ds),"Test:",len(test_ds))
class_names=[name for name,index in sorted(train_raw.class_to_idx.items(),key=lambda x:x[1])]
def build_resnet():
    model=models.resnet50(weights=models.ResNet50_Weights.DEFAULT)
    for p in model.parameters(): p.requires_grad=False
    model.fc=nn.Linear(model.fc.in_features,NUM_CLASSES)
    return model
def build_vit():
    model=models.vit_b_16(weights=models.ViT_B_16_Weights.DEFAULT)
    for p in model.parameters(): p.requires_grad=False
    model.heads.head=nn.Linear(model.heads.head.in_features,NUM_CLASSES)
    return model
```

```
def run_epoch(model,loader,criterion,optimizer=None):
    training=optimizer is not None
    model.train() if training else model.eval()
    total_loss=0
    correct=0
    total=0
    all_preds=[]
    all_labels=[]
    with torch.set_grad_enabled(training):
        for images,labels in loader:
            images=images.to(DEVICE,non_blocking=True)
            labels=labels.to(DEVICE,non_blocking=True)
            if training: optimizer.zero_grad(set_to_none=True)
            outputs=model(images)
            loss=criterion(outputs,labels)
            if training:
                loss.backward()
                optimizer.step()
            total_loss+=loss.item()*labels.size(0)
            preds=outputs.argmax(1)
            correct+=(preds==labels).sum().item()
            total+=labels.size(0)
            all_preds.extend(preds.cpu().numpy())
            all_labels.extend(labels.cpu().numpy())
    return total_loss/total,correct/total,all_labels,all_preds
```

```python
def run_model(name,builder):
    print("\n"+"="*65+"\nMODEL:",name+"\n"+"="*65)
    start=time.time()
    model=builder().to(DEVICE)
    total_params=sum(p.numel() for p in model.parameters())
    trainable_params=sum(p.numel() for p in model.parameters() if p.requires_grad)
    print("Total parameters:",f"{total_params:,}")
    print("Trainable parameters:",f"{trainable_params:,}")
    criterion=nn.CrossEntropyLoss()
    optimizer=optim.Adam((p for p in model.parameters() if p.requires_grad),lr=LEARNING_RATE)
    history={"train_loss":[],"train_acc":[],"val_loss":[],"val_acc":[]}
    for epoch in range(EPOCHS):
        epoch_start=time.time()
        train_loss,train_acc,_,_=run_epoch(model,train_loader,criterion,optimizer)
        val_loss,val_acc,_,_=run_epoch(model,val_loader,criterion)
        history["train_loss"].append(train_loss)
        history["train_acc"].append(train_acc)
        history["val_loss"].append(val_loss)
        history["val_acc"].append(val_acc)
```

```python
        print(f"Epoch {epoch+1}/{EPOCHS} | Train Loss: {train_loss:.4f} | Train Acc: {train_acc*100:.2f}% | Val Loss: {val_loss:.4f} | Val Acc: {val_acc*100:.2f}% | Time: {(time.tim
    test_loss,test_acc,y_true,y_pred=run_epoch(model,test_loader,criterion)
    result={"model":name,"accuracy":accuracy_score(y_true,y_pred),"precision":precision_score(y_true,y_pred,average="macro",zero_division=0),"recall":recall_score(y_true,y_pred,aver
    print("\nTEST RESULTS")
    print(f"Test Loss: {test_loss:.4f}")
    print(f"Accuracy: {result['accuracy']*100:.2f}%")
    print(f"Macro Precision: {result['precision']*100:.2f}%")
    print(f"Macro Recall: {result['recall']*100:.2f}%")
    print(f"Macro F1: {result['f1']*100:.2f}%")
    print(f"Total model time: {result['time_min']:.2f} minutes")
    del model,optimizer,criterion
    gc.collect()
    if torch.cuda.is_available(): torch.cuda.empty_cache()
    return result
results=[]
results.append(run_model("ResNet50 CNN",build_resnet))
results.append(run_model("ViT-B/16",build_vit))
print("\n"+"="*65+"\nFINAL COMPARISON\n"+"="*65)
for r in results:
    print(f"{r['model']}: Accuracy={r['accuracy']*100:.2f}% | Precision={r['precision']*100:.2f}% | Recall={r['recall']*100:.2f}% | F1={r['f1']*100:.2f}% | Time={r['time_min']:.2f}
metrics=["accuracy","precision","recall","f1"]
x=np.arange(len(metrics))
width=0.35
plt.figure(figsize=(9,5))
for i,r in enumerate(results):
    plt.bar(x+(i-0.5)*width,[r[m]*100 for m in metrics],width,label=r["model"])
plt.xticks(x,["Accuracy","Macro Precision","Macro Recall","Macro F1"])
plt.ylabel("Score (%)")
plt.ylim(0,100)
plt.title("ViT vs ResNet50: Classification Metrics")
plt.legend()
plt.tight_layout()
plt.show()
```

```python
for ax,r in zip(axes,results):
    h=r["history"]
    ax.plot(range(1,EPOCHS+1),[v*100 for v in h["train_acc"]],marker="o",label="Train")
    ax.plot(range(1,EPOCHS+1),[v*100 for v in h["val_acc"]],marker="o",label="Validation")
    ax.set_title(r["model"])
    ax.set_xlabel("Epoch")
    ax.set_ylabel("Accuracy (%)")
    ax.legend()
plt.tight_layout()
plt.show()
fig,axes=plt.subplots(1,2,figsize=(17,7))
for ax,r in zip(axes,results):
    cm=confusion_matrix(r["y_true"],r["y_pred"],labels=list(range(NUM_CLASSES)))
    sns.heatmap(cm,ax=ax,cmap="Blues",xticklabels=class_names,yticklabels=class_names,cbar=False)
    ax.set_title(r["model"]+" Confusion Matrix")
    ax.set_xlabel("Predicted")
    ax.set_ylabel("Actual")
    ax.tick_params(axis="x",labelrotation=90,labelsize=6)
    ax.tick_params(axis="y",labelsize=6)
plt.tight_layout()
plt.show()
for r in results:
    print("\nClassification report:",r["model"])
    print(classification_report(r["y_true"],r["y_pred"],labels=list(range(NUM_CLASSES)),target_names=class_names,zero_division=0))
print("\nTotal experiment time:",f"{sum(r['time_min'] for r in results):.2f} minutes")
```

Output :

```
============================================================
MODEL: ViT-B/16
============================================================
Downloading: "https://download.pytorch.org/models/vit_b_16-c867db91.pth" to /root/.cache/torch/hub/checkpoints/vit_b_16-c867db91.pt
100%|██████████| 330M/330M [00:02<00:00, 138MB/s]
Total parameters: 85,827,109
Trainable parameters: 28,453
Epoch 1/2 | Train Loss: 2.8799 | Train Acc: 45.41% | Val Loss: 1.9559 | Val Acc: 81.62% | Time: 0.19 min
Epoch 2/2 | Train Loss: 1.3333 | Train Acc: 87.30% | Val Loss: 0.9749 | Val Acc: 87.57% | Time: 0.19 min

TEST RESULTS
Test Loss: 0.9479
Accuracy: 88.11%
Macro Precision: 90.68%
Macro Recall: 88.11%
Macro F1: 87.69%
Total model time: 0.50 minutes

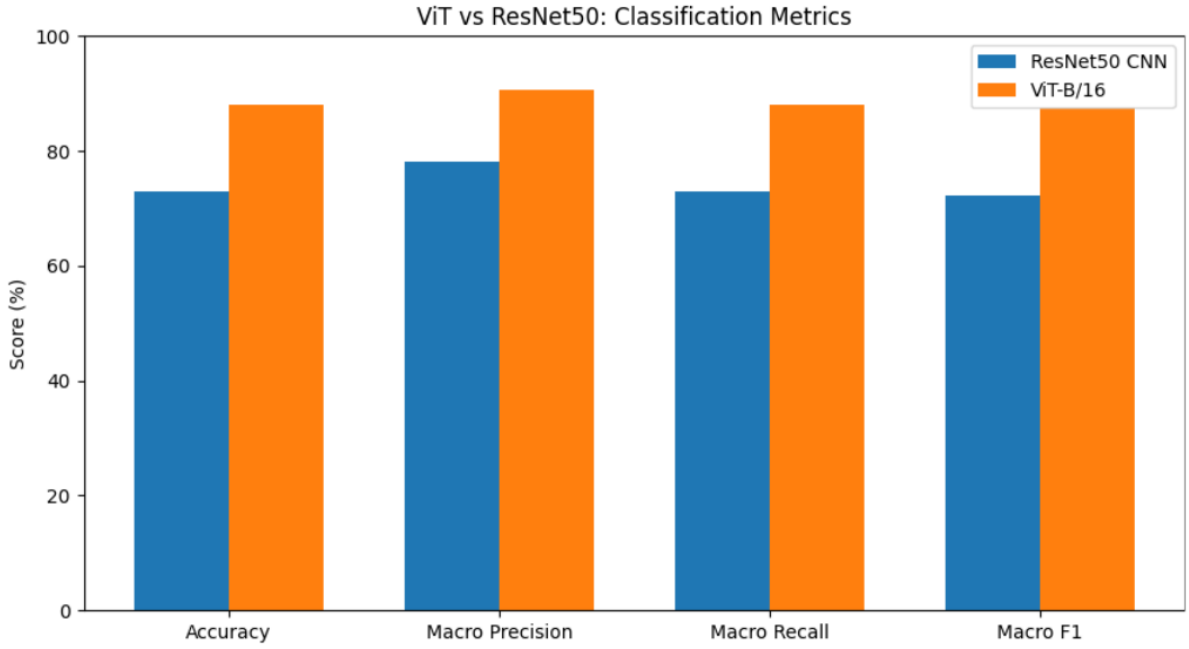
============================================================
FINAL COMPARISON
============================================================
ResNet50 CNN: Accuracy=72.97% | Precision=78.25% | Recall=72.97% | F1=72.34% | Time=0.26 min
ViT-B/16: Accuracy=88.11% | Precision=90.68% | Recall=88.11% | F1=87.69% | Time=0.50 min
```

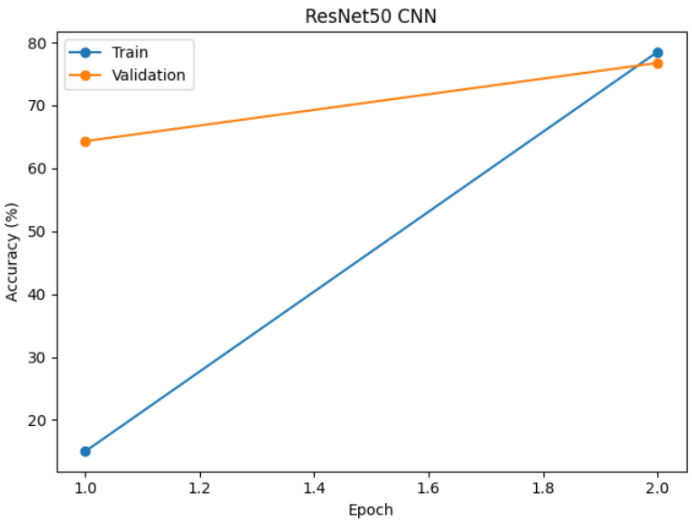
```
Device: cuda
GPU: Tesla T4
100%|██████████| 792M/792M [00:30<00:00, 25.9MB/s]
100%|██████████| 19.2M/19.2M [00:01<00:00, 13.8MB/s]
Classes: 37
Train: 740 Validation: 185 Test: 185

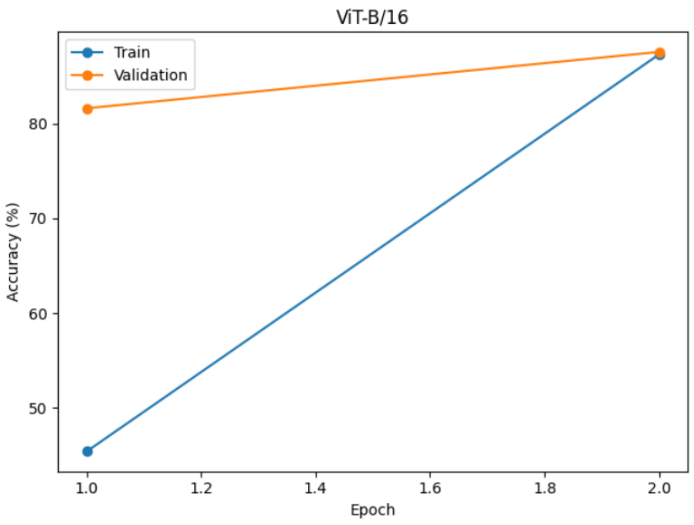
============================================================
MODEL: ResNet50 CNN
============================================================
Downloading: "https://download.pytorch.org/models/resnet50-11ad3fa6.pth" to /root/.cache/torch/hub/checkpoints/resnet50-11ad3fa6.pth
100%|██████████| 97.8M/97.8M [00:00<00:00, 157MB/s]
Total parameters: 23,583,845
Trainable parameters: 75,813
Epoch 1/2 | Train Loss: 3.3606 | Train Acc: 15.00% | Val Loss: 2.9760 | Val Acc: 64.32% | Time: 0.13 min
Epoch 2/2 | Train Loss: 2.4965 | Train Acc: 78.51% | Val Loss: 2.4172 | Val Acc: 76.76% | Time: 0.09 min

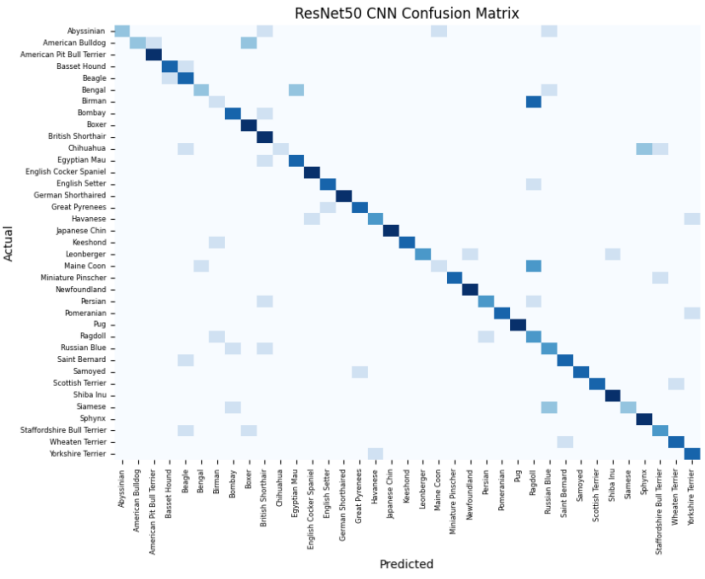
TEST RESULTS
Test Loss: 2.4497
Accuracy: 72.97%
Macro Precision: 78.25%
Macro Recall: 72.97%
Macro F1: 72.34%
Total model time: 0.26 minutes
```

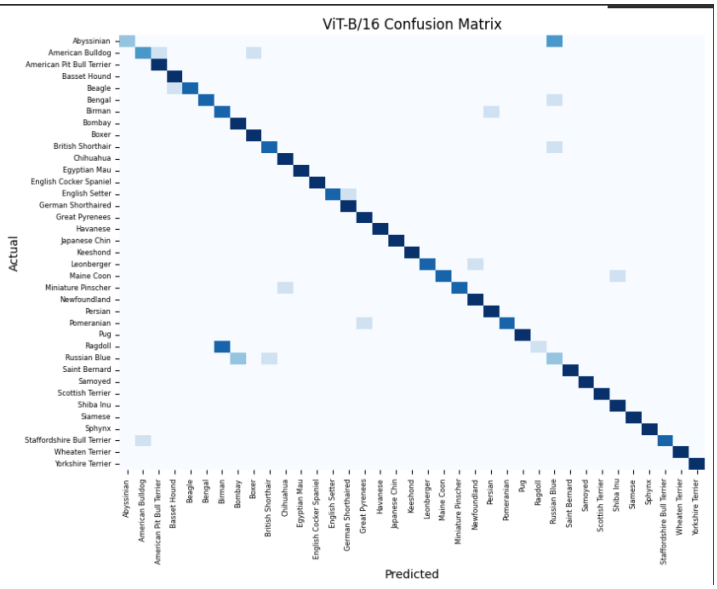
```
============================================================
FINAL COMPARISON
============================================================
ResNet50 CNN: Accuracy=72.97% | Precision=78.25% | Recall=72.97% | F1=72.34% | Time=0.26 min
ViT-B/16: Accuracy=88.11% | Precision=90.68% | Recall=88.11% | F1=87.69% | Time=0.50 min
```











Classification report: ResNet50 CNN

	precision	recall	f1-score	support
Abyssinian	1.00	0.40	0.57	5
American Bulldog	1.00	0.40	0.57	5
American Pit Bull Terrier	0.83	1.00	0.91	5
Basset Hound	0.80	0.80	0.80	5
Beagle	0.50	0.80	0.62	5
Bengal	0.67	0.40	0.50	5
Birman	0.33	0.20	0.25	5
Bombay	0.67	0.80	0.73	5
Boxer	0.62	1.00	0.77	5
British Shorthair	0.50	1.00	0.67	5
Chihuahua	1.00	0.20	0.33	5
Egyptian Mau	0.67	0.80	0.73	5
English Cocker Spaniel	0.83	1.00	0.91	5
English Setter	0.80	0.80	0.80	5
German Shorthaired	1.00	1.00	1.00	5
Great Pyrenees	0.80	0.80	0.80	5
Havanese	0.75	0.60	0.67	5
Japanese Chin	1.00	1.00	1.00	5
Keeshond	1.00	0.80	0.89	5
Leonberger	1.00	0.60	0.75	5
Maine Coon	0.50	0.20	0.29	5
Miniature Pinscher	1.00	0.80	0.89	5
Newfoundland	0.83	1.00	0.91	5
Persian	0.75	0.60	0.67	5
Pomeranian	1.00	0.80	0.89	5
Pug	1.00	1.00	1.00	5
Ragdoll	0.25	0.60	0.35	5
Russian Blue	0.43	0.60	0.50	5
Saint Bernard	0.80	0.80	0.80	5
Samoyed	1.00	0.80	0.89	5
Scottish Terrier	1.00	0.80	0.89	5
Shiba Inu	0.83	1.00	0.91	5
Siamese	1.00	0.40	0.57	5
Sphynx	0.71	1.00	0.83	5
Staffordshire Bull Terrier	0.60	0.60	0.60	5
Wheaten Terrier	0.80	0.80	0.80	5
Yorkshire Terrier	0.67	0.80	0.73	5
accuracy			0.73	185
macro avg	0.78	0.73	0.72	185
weighted avg	0.78	0.73	0.72	185

Classification report: ViT-B/16

	precision	recall	f1-score	support
Abyssinian	1.00	0.40	0.57	5
American Bulldog	0.75	0.60	0.67	5
American Pit Bull Terrier	0.83	1.00	0.91	5
Basset Hound	0.83	1.00	0.91	5
Beagle	1.00	0.80	0.89	5
Bengal	1.00	0.80	0.89	5
Birman	0.50	0.80	0.62	5
Bombay	0.71	1.00	0.83	5
Boxer	0.83	1.00	0.91	5
British Shorthair	0.80	0.80	0.80	5
Chihuahua	0.83	1.00	0.91	5
Egyptian Mau	1.00	1.00	1.00	5
English Cocker Spaniel	1.00	1.00	1.00	5
English Setter	1.00	0.80	0.89	5
German Shorthaired	0.83	1.00	0.91	5
Great Pyrenees	0.83	1.00	0.91	5
Havanese	1.00	1.00	1.00	5
Japanese Chin	1.00	1.00	1.00	5
Keeshond	1.00	1.00	1.00	5
Leonberger	1.00	0.80	0.89	5
Maine Coon	1.00	0.80	0.89	5
Miniature Pinscher	1.00	0.80	0.89	5
Newfoundland	0.83	1.00	0.91	5
Persian	0.83	1.00	0.91	5
Pomeranian	1.00	0.80	0.89	5

Keeshond	1.00	1.00	1.00	5
Leonberger	1.00	0.80	0.89	5
Maine Coon	1.00	0.80	0.89	5
Miniature Pinscher	1.00	0.80	0.89	5
Newfoundland	0.83	1.00	0.91	5
Persian	0.83	1.00	0.91	5
Pomeranian	1.00	0.80	0.89	5
Pug	1.00	1.00	1.00	5
Ragdoll	1.00	0.20	0.33	5
Russian Blue	0.29	0.40	0.33	5
Saint Bernard	1.00	1.00	1.00	5
Samoyed	1.00	1.00	1.00	5